

Catalog Scheduling Statistical Results

About this project

My environment is a lot like cliff walker project.

When running the environment you can choose “catalog-measure” or “catalog-search”.

Catalog-search: here is where we use random restart hill climbing to find the catalog with the best utility. For this part of my statistical results, I will be changing the number of random restarts for hill climbing and the number of students for the utility function since it is a simulator so the more students the better.

Catalog-measure: the agents take a catalog that has been searched before and simulate a student taking classes. The reward is the semesters that student took to graduate. The more times you run the agent the more accurate the results will be since it returns the average semesters of all the students.

Catalog Search

For each of these runs the maximum credits per semester are 15.

Restarts	Students	Utility (semesters)	Run Time (seconds)
5	100	9.63	5.80
10	100	9.68	11.15
50	100	9.69	56.30
5	1,000	9.74	50.06
10	1,000	9.70	114.44
50	1,000	9.74	565.90
10	10,000	9.72	1105.58

Random Agent

The random agent takes the observation and randomly chooses classes from the ones that are available until the maximum credit per semester is met.

For this I will always be using the catalog found from 10,000 students above.

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	15.00	12	20	100
15	12.00	10	17	100
18	11.00	8	14	100
20	10.00	7	14	100

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	14.00	10	21	1,000
15	12.00	9	18	1,000
18	11.00	7	16	1,000
20	10.00	8	15	1,000

Requirements Agent

This agent also takes the observation but instead of taking random classes it takes classes based on the requirements to graduate. For example, if all the elective credits are met then there is no reason to take more elective classes.

For this I will always be using the catalog found from 10,000 students above.

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	8.00	7	10	100
15	6.00	6	8	100
18	5.00	5	6	100
20	5.00	5	6	100

Credits	Av. Semesters	Min Semester	Max Semester	Students
12	7.00	7	10	1,000
15	6.00	6	8	1,000
18	5.00	5	6	1,000
20	5.00	5	6	1,000

Annotations

We need to have into account that the generals are not part of the catalog so it wouldn't take 8 semesters to graduate.

The random student simulator makes the utility a little different every time since each student will be taking random classes and that can make the same catalog have different utilities so there is a little of error there sometimes.